Tutorial – 4

1 : The employee list for a company contains employee code, name, designation and basic pay. The employee is given a house rent allowance (HRA) of 10% of the basic pay and dearness allowance (DA) of 45% of the basic pay. The total pay of the employee is calculated as Basic Pay + HRA + DA. Write a class to define the details of the employee. Write a constructor to assign the required initial values. Add a method to calculate HRA, DA and total pay and print them. Write another class with main method. Create objects for three different employees and calculate HRA, DA and total pay.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_lunagariya_rajvi_Tutorial_4
{
   class Employee
   {
      private int emp_code;
      private string emp_name;
      private string designation;
      private double basicPay;

      public Employee(int emp_code, string emp_name, string designation, double
basicPay)
      {
         this.emp_code = emp_code;
         this.emp_name = emp_name;
         this.designation = designation;
         this.basicPay = basicPay;
      }
```

```
    public void CalculatePay()
    {
        double hra = basicPay * 0.10;
        double da = basicPay * 0.45;
        double netSalary = basicPay + hra + da;

        Console.WriteLine("Employee Code: " + emp_code);
        Console.WriteLine("Employee Name: " + emp_name);
        Console.WriteLine("Designation: " + designation);
        Console.WriteLine("Basic Pay: " + basicPay);
        Console.WriteLine("HRA: " + hra);
        Console.WriteLine("DA: " + da);
        Console.WriteLine("Gross Pay: " + netSalary);
    }


}

class Employee1
{
    public static void Main()
    {

        Employee e1 = new Employee(101, "MAHI", "Software Engineer",
50000);
        e1.CalculatePay();

        Employee e2 = new Employee(101, "rajvu", " backend devloper", 50000);
        e2.CalculatePay();

        Employee e3 = new Employee(101, "krisha", "computer engineering",
50000);
        e3.CalculatePay();
```

```
            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>Employee.exe
Employee Code: 101
Employee Name: MAHI
Designation: Software Engineer
Basic Pay: 50000
HRA: 5000
DA: 22500
Gross Pay: 77500
Employee Code: 101
Employee Name: rajvu
Designation:  backend devloper
Basic Pay: 50000
HRA: 5000
DA: 22500
Gross Pay: 77500
Employee Code: 101
Employee Name: krisha
Designation: computer engineering
Basic Pay: 50000
HRA: 5000
DA: 22500
Gross Pay: 77500
```

2 : From the following code and given output, complete missing statements and find out error code and correct it.

Solution:

```
//Base class or Parent class.
class Shape
{
   public double Width;
   public double Height;
   public void ShowDim()
   {
      Console.WriteLine("Width and height are " +  Width + " and " +
Height);
```

```
    }
}
// Triangle is derived from Shape.
//Drived class or Child class.
class Triangle : Shape
{
   public string Style; // style of triangle
   // Return area of triangle.
 //……Missing statement-
1……..//
   {
      return Width * Height / 2;
   }
   // Display a triangle's style.
   public void ShowStyle()
   {
//……Missing statement-
2……..//
   }
}
//Driver class which runs the program.
class Driver
{
   static void Main()
   {
      Triangle t1 new Triangle();
      Triangle t2 new Triangle();

      t1.Width =4.0;
      t1.Height =4.0;
      t1.Style ="isosceles";
      t2.Width =8.0;
      t2.Height =12.0;
      t2.Style ="right";

      Console.WriteLine("Info for t1: ");
      t1.ShowStyle();
```

```
        t1.ShowDim();
        Console.WriteLine("Area is " + t1.Area());

        Console.WriteLine();

        Console.WriteLine("Info for t2: ");
        t2.ShowStyle();
        t2.ShowDim();
        Console.WriteLine("Area is " + t2.Area());
    }
}
```

Output

```
Info for t1:
Triangle is isosceles
Width and height are 4 and 4
Area is 8

Info for t2:
Triangle is right
Width and height are 8 and 12
Area is 48
```

```
using System;

class shpae
{
    public double Width;
    public double Height;

    public void ShowDim()
    {
        Console.WriteLine("Width and height are " + Width + " and " + Height);
    }
}

// Triangle is derived from Shape.
// Derived class or Child class.
```

```
class Triangle : shpae
{
   public string Style; // style of triangle

   // Return area of triangle.
   public double Area()
   {
      return Width * Height / 2;
   }

   // Display a triangle's style.
   public void ShowStyle()
   {
      Console.WriteLine("Triangle is " + Style);
   }
}

// Driver class which runs the program.
class shape
{
   static void Main()
   {
      Triangle t1 = new Triangle();
      Triangle t2 = new Triangle();

      t1.Width = 4.0;
      t1.Height = 4.0;
      t1.Style = "isosceles";

      t2.Width = 8.0;
      t2.Height = 12.0;
      t2.Style = "right";

      Console.WriteLine("Info for t1: ");
      t1.ShowStyle();
      t1.ShowDim();
      Console.WriteLine("Area is " + t1.Area());
```

```
        Console.WriteLine();

        Console.WriteLine("Info for t2: ");
        t2.ShowStyle();
        t2.ShowDim();
        Console.WriteLine("Area is " + t2.Area());
    }
}
```

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>shpae.exe
Info for t1:
Triangle is isosceles
Width and height are 4 and 4
Area is 8

Info for t2:
Triangle is right
Width and height are 8 and 12
Area is 48
```

3: Draw a real picture for single level inheritance.
Perform following tasks.
Task 1: Create a class
Task 2: Add few data members as private, protected and public
Task 3: Add few methods as public to work on defined data members
Task 4: Create another applicable class which inherits members from above class
Task 5: Add few data members as private, protected and public into second class
Task 6: Add few methods as public to work on defined data members into second class
Task 7: Create a Demo class with main method.
Task 8: Create at least two objects of a second class defined in Task 4 into main method and call all methods using that object.
Task 9: Write comment for each important portion of code like data members' declaration, methods, some important logic etc.
Task 10: Summarize above solution in your own few words to visualize the solution to the end user.

```
class Collage
 {
    private string name;
    protected string uni_name;
    public string department;

    public Collage(string name, string uni_name, string department)
    {
       this.name = name;
       this.uni_name = uni_name;
       this.department = department;
    }

    public void DisplayName()
    {
       Console.WriteLine("Collage Name: " + name);

    }

    public void DisplayUniName()
    {
       Console.WriteLine("university Name: " + uni_name);

    }

    public void DisplayDepName()
    {
       Console.WriteLine("Department Name: " + department);

    }

    public void Display()
    {
       DisplayName();
       DisplayUniName();
       DisplayDepName();
```

```
        }

        //child class
        class SOE : Collage
        {
            private string Event_name;
            protected string Cost_of_event;
            public string co_ordinator_name;

            public SOE(string name, string uni_name, string department, string
Event_name, string Cost_of_event, string co_ordinator_name)
                : base(name, uni_name, department)
            {
                this.Event_name = Event_name;
                this.Cost_of_event = Cost_of_event;
                this.co_ordinator_name = co_ordinator_name;
            }

            public void DisplayEventName()
            {
                Console.WriteLine("Event Name: " + Event_name);
            }

            public void DisplayCostOfEvent()
            {
                Console.WriteLine("Cost of Event: " + Cost_of_event);
            }

            public void DisplayCoordinatorName()
            {
                Console.WriteLine("Co-ordinator Name: " + co_ordinator_name);
            }

            public void DisplaySOE()
            {
                Display();
                DisplayEventName();
```

```
            DisplayCostOfEvent();
            DisplayCoordinatorName();
        }

    }


    class T4_program3
    {
        static void Main()
        {
            SOE s1 = new SOE("Global", "RKU", "Computer science",
"Techno_planet", "50000", "Raviraj");
            s1.Display();
            s1.DisplaySOE();

            SOE s2 = new SOE("Global", "RKU", "Computer science",
"Techno_planet", "50000", "Raviraj");
            s2.Display();
            s2.DisplaySOE();

            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>Collage.exe
Collage Name: Global
University Name: RKU
Department Name: Computer science
Collage Name: Global
University Name: RKU
Department Name: Computer science
Event Name: Techno_planet
Cost of Event: 50000
Co-ordinator Name: Raviraj
Collage Name: Global
University Name: RKU
Department Name: Computer science
Collage Name: Global
University Name: RKU
Department Name: Computer science
Event Name: Techno_planet
Cost of Event: 50000
Co-ordinator Name: Raviraj
```

4 : From the following code and given output complete missing statements and find out error code and correct it.

```
using System;
namespace StaticVarApplication
{
   class StaticVar
   {
     public static int num;

      public counting()
      {
         num++;
      }
//……Missing statement……..//
      {
         return num;
      }
   }
```

```
    class StaticTester
    {
        static void Main(string[] args)
        {
            StaticVar s = new StaticVar();
            s.count();
            s.count();
            s.count();
            Console.WriteLine("Variable num: {0}", StaticVar.getNum());
            Console.ReadKey();
        }
    }
}
```

Output:

Variable num: 3

```
using System;
namespace StaticVarApplication
{
    class StaticVar
    {
      public static int num;

        public counting()
        {
            num++;
        }
//……Missing
statement……..//
        {
            return num;
        }
    }
    class StaticTester
    {
        static void Main(string[] args)
        {
```

```
            StaticVar s = new StaticVar();
            s.count();
            s.count();
            s.count();
            Console.WriteLine("Variable num: {0}", StaticVar.getNum());
            Console.ReadKey();
        }
    }
}
```

Output:
Variable num: 3

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>StaticVar.exe
Variable num: 3
```

5 : Find out error code and correct it. Print appropriate output as desired.

```
using System;
public class A // This is the base class.
{
   public A(int value)
   {
    // Executes some code in the constructor.
    Console.WriteLine("Base constructor A()");
   }
}

public class B : A // This class derives from the previous class.
{
   public B(int value) : base(value)
   {
    // The base constructor is called first.
    //... Then this code is executed.
    Console.WriteLine("Derived constructor B()");
   }
}
```

```
class Program
{
   static void Main()
   {
    // Create a new instance of class A, which is the base class.
    // ... Then create an instance of B, which executes the base constructor.
     A a = A(0);
     B b = B(1);
   }
}
```

Expected Outcome :

```
Base constructor A()
Base constructor A()
Derived constructor B()
```

```
using System;

namespace _24SOECE11019_lunagariya_rajvi_Tutorial_4
{
   // Base class
   public class A
   {
      // Constructor of base class
      public A(int value)
      {
         Console.WriteLine("Base constructor A()");
      }
   }

   // Derived class
   public class B : A
   {
      // Constructor of derived class
      // base(value) calls the constructor of class A
      public B(int value) : base(value)
      {
         Console.WriteLine("Derived constructor B()");
```

```
        }
    }

    // Main/Driver class
    class Program
    {
        static void Main()
        {
            // Create object of base class A
            A a = new A(0);

            B b = new B(1);

            Console.ReadKey();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>A.exe
Base constructor A()
Base constructor A()
Derived constructor B()
^C
```

6 : Find out error code and correct it. Print appropriate output as desired.
Solution:

```
using System;
abstract class Test
{
    int a;
    abstract void A();
}

class Example1 : Test
```

```
{
   public override void A()
   {
    Console.WriteLine("Example1.A");
    base.a++;
   }
}
class Example2 : Test
{
   public override void A()
   {
    Console.WriteLine("Example2.A");
    base.a--;
   }
}
class Program
{
   static void Main()
   {
    // Reference Example1 through Test type.
    Test test1 = new Example1();
    test1.A();
    // Reference Example2 through Test type.
    Test test2 = new Example2();
    test2.A();
   }
}
```

Expected Outcome :

Example1.A
Example2.A

```
using System;
abstract class Test
{
   int a;
   abstract void A();
}
```

```
class Example1 : Test
{
   public override void A()
    {
     Console.WriteLine("Example1.A");
     base.a++;
    }
}
class Example2 : Test
{
   public override void A()
    {
     Console.WriteLine("Example2.A");
     base.a--;
    }
}
class Program
{
   static void Main()
    {
     // Reference Example1 through Test type.
     Test test1 = new Example1();
     test1.A();
     // Reference Example2 through Test type.
     Test test2 = new Example2();
     test2.A();
    }
}
```

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>Test.exe
Example1.A
Example2.A
```

7 : Refer given output and find out error code and correct it.

```
sealed class A
{
```

```
        public int x;
        public int y;
}
class B : A
{
        public int z;
}
class SealedTest2
{
        static void Main()
        {
           A sc = new A();
           sc.x = 110;
           sc.y = 150;
           Console.WriteLine("x = {0}, y = {1}", sc.x, sc.y);
        }
}
```

Output

x = 110, y = 150

```
sealed class A
{
        public int x;
        public int y;
}
class B : A
{
        public int z;
}
class SealedTest2
{
        static void Main()
        {
           A sc = new A();
           sc.x = 110;
           sc.y = 150;
           Console.WriteLine("x = {0}, y = {1}", sc.x, sc.y);
```

```
        }
}
```

Output

x = 110, y = 150

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>A1.exe
x = 110, y = 150
```

8 : Find out error code and correct it. Print appropriate output as desired.

Solution:

```
class X
 {
    protected virtual void F() { Console.WriteLine("X.F"); }
    protected virtual void F2() { Console.WriteLine("X.F2"); }
 }
 class Y : X
 {
    sealed protected override void F() { Console.WriteLine("Y.F"); }
    protected override void F2() { Console.WriteLine("Y.F2"); }
 }
 class Z : Y
 {
    // Overriding F
    protected override void F() { Console.WriteLine("Z.F"); }

    // Overriding F2
    protected override void F2() { Console.WriteLine("Z.F2"); }
 }
class SealedMethodTest
{
    static void Main()
    {
       X Obj1 = new X();
       Obj1.F();
       Obj2.F2();
```

```
            Y Obj2 = new Y();
            Obj2.F();
            Obj2.F2();

            Z Obj3 = new Z();
            Obj3.F();
            Obj3.F2();
        }
    }
```

Expected Outcome :

```
 X.F
X.F2
Y.F
Y.F2
Y.F
Z.F2
```

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_lunagariya_rajvi_Tutorial_4
{
    class X
    {
        // Virtual methods in base class
        public virtual void F()
        {
            Console.WriteLine("X.F");
        }

        public virtual void F2()
        {
```

```
        Console.WriteLine("X.F2");
    }
}

class Y : X
{
    // Sealed method cannot be overridden further
    public sealed override void F()
    {
        Console.WriteLine("Y.F");
    }

    // F2 is not sealed, so it can be overridden by Z
    public override void F2()
    {
        Console.WriteLine("Y.F2");
    }
}

class Z : Y
{
    // We CANNOT override F here because Y.F() is sealed.

    // Overriding F2
    public override void F2()
    {
        Console.WriteLine("Z.F2");
    }
}

class SealedMethodTest
{
    static void Main()
    {
        // Object of X
        X Obj1 = new X();
```

```
            Obj1.F();
            Obj1.F2();

            // Object of Y
            Y Obj2 = new Y();

            Obj2.F();
            Obj2.F2();

            // Object of Z
            Z Obj3 = new Z();

            // F() comes from Y because it is sealed in Y
            Obj3.F();

            // F2() is overridden in Z
            Obj3.F2();

            Console.ReadKey();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>X.exe
X.F
X.F2
Y.F
Y.F2
Y.F
Z.F2
^C
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>
```

9 : This program will throw an exception. Add try, catch and finally blocks to handle this exception.

```
using System;
class MyClient
{
    public static void Main()
```

```
        {
            int x = 0;
            int div = 100/x;
            Console.WriteLine(div);
        }
}
```

Expected Outcome :
Cannot divide by zero.
Finally block executed.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_lunagariya_rajvi_Tutorial_4
{


   class MyClient
   {
      public static void Main()
      {
         try
         {
            int x = 0;

            // This causes DivideByZeroException
            int div = 100 / x;

            Console.WriteLine(div);
         }
         catch (DivideByZeroException)
         {
            // Handle divide-by-zero error
```

```
            Console.WriteLine("Cannot divide by zero.");
        }
        finally
        {
            // This block always executes
            Console.WriteLine("Finally block executed.");
        }

        Console.ReadKey();
    }
  }
}
```

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>Myclient.exe
Cannot divide by zero.
Finally block executed.
```

10: Arrange the code to get desirable output

```
class MyException : Exception
{
}
class MyClient
{
   public static void Main()
   {
       try
       {
            throw new MyException("my exception generated.");
       }
       catch(Exception e)
       {

       }
       Console.WriteLine("LAST STATEMENT");
   }
            using System;
```

```
            public MyException(string str)
    {
            Console.WriteLine("User defined exception");
            Console.WriteLine("Exception caught here: " + e.ToString());
              }

}
```

Output:
Exception caught here: my exception generated.
LAST STATEMENT

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_lunagariya_rajvi_Tutorial_4
{


    // User-defined exception class
    class Mxception : Exception
    {
        // Constructor of custom exception
        public Mxception(string str) : base(str)
        {
            Console.WriteLine("User defined exception");
        }
    }

    class MyClient1
    {
        public static void Main()
        {
            try
```

```
            {
                // Throw user-defined exception
                throw new Mxception("my exception generated.");
            }
            catch (Exception e)
            {
                // Display the exception message
                Console.WriteLine("Exception caught here: " + e.Message);
            }

            // This statement executes after the exception is handled
            Console.WriteLine("LAST STATEMENT");

            Console.ReadKey();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>M
User defined exception
Exception caught here: my exception generated.
LAST STATEMENT
^C
D:\.net Project\24SOECE11019_lunagariya_rajvi_Tutorial-4>
```